

28. Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

a. Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?

Rewriting the demand equation with price on the left-hand side:

$$\begin{aligned} Q_i &= \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di} \\ \alpha P_i &= Q_i - \alpha_1 - \alpha_3 PS_i - \alpha_4 DI_i - e_{di} \\ P_i &= \frac{1}{\alpha_2} Q_i - \frac{\alpha_1}{\alpha_2} - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i - \frac{1}{\alpha_2} e_{di} \\ P_i &= \gamma_1 + \gamma_2 Q_i + \gamma_3 PS_i + \gamma_4 DI_i + u_{di} \end{aligned}$$

where $\gamma_2 = \frac{1}{\alpha_2} < 0$, $\gamma_3 = -\frac{\alpha_3}{\alpha_2} > 0$, and $\gamma_4 = -\frac{\alpha_4}{\alpha_2} > 0$.

Rewriting the supply equation with price on the left-hand side:

$$\begin{aligned} Q_i &= \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si} \\ \beta_2 P_i &= Q_i - \beta_1 - \beta_3 PF_i - e_{si} \\ P_i &= \frac{1}{\beta_2} Q_i - \frac{\beta_1}{\beta_2} - \frac{\beta_3}{\beta_2} PF_i - \frac{1}{\beta_2} e_{si} \\ P_i &= \delta_1 + \delta_2 Q_i + \delta_3 PF_i + u_{si} \end{aligned}$$

where $\delta_2 = \frac{1}{\beta_2} > 0$ and $\delta_3 = -\frac{\beta_3}{\beta_2} > 0$.

b. Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?

We estimate the supply and demand equations using two-stage least squares (2SLS), treating Q as endogenous. For the demand equation, PF is used as an instrument for Q . For the supply equation, PS and DI are used as instruments for Q .

In the demand equation, the coefficient on Q is -2.671 and is statistically significant at the 5% level ($p = 0.0315$), indicating a downward-sloping demand curve. The coefficients on PS (3.461) and DI (13.390) are positive and statistically significant, as expected.

	coefficient	p-value
Q	-2.671	0.03154*
PS	3.461	0.00458**
DI	13.390	4.68×10^{-5} ***

In the supply equation, the coefficient on Q is 2.937 and highly significant ($p < 0.001$), indicating an upward-sloping supply curve. The coefficient on PF (2.959) is also positive and highly significant, consistent with theory.

	coefficient	p-value
Q	2.9367	$1.32 \times 10^{-13}***$
PF	2.9585	$< 2 \times 10^{-16}***$

Overall, the estimated coefficients have the expected signs and are statistically significant, supporting the validity of the model and consistency with economic theory.

c. Estimate the price elasticity of demand “at the means” using the results from (b).

The price elasticity of demand is computed as:

$$Elasticity = \frac{dQ}{dP} \cdot \frac{P}{Q}$$

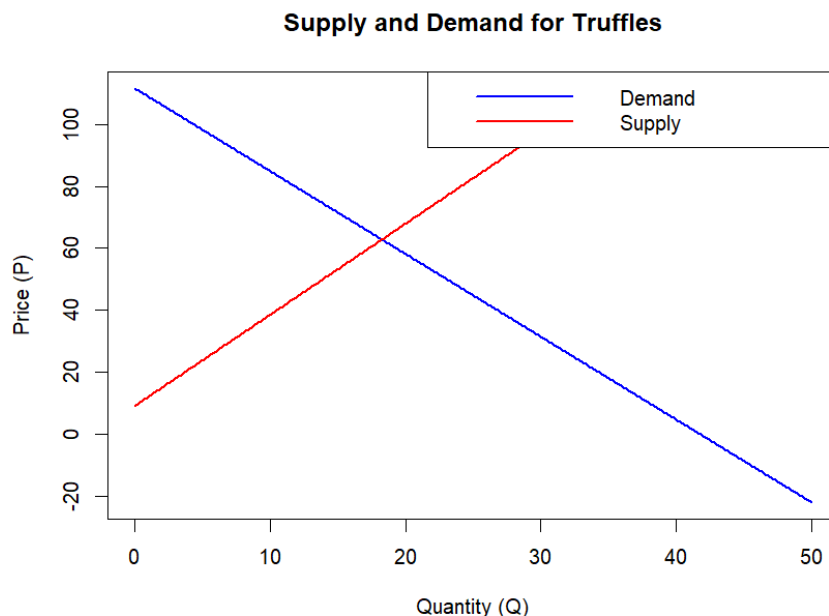
$$\frac{dP}{dQ} = \gamma_2$$

$$E = (1 / \gamma_2) \times (\bar{P} / \bar{Q})$$

Using the estimated coefficient $\gamma_2 = -2.671$ and the sample means of price and quantity, we obtain a negative elasticity, consistent with the law of demand.

The price elasticity of demand at the sample means is -1.272. This value is negative, which is consistent with the law of demand. Since the absolute value of the elasticity is greater than one, demand is elastic at the mean.

d. Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI, PS , and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.



Using the estimated equations and setting $DI = 3.5$, $PS = 22$, and $PF = 23$, we construct the demand and supply curves. The demand curve is downward sloping, and the supply curve is upward sloping. The intersection of the two curves determines the equilibrium price and quantity.

e. What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?

$$\begin{cases} P = -11.428 - 2.671 * Q + 3.461 * PS + 13.390 * DI \\ P = -58.7982 + 2.9367 * Q + 2.9585 * PF \end{cases}$$

$$\begin{cases} P = -11.428 - 2.671 * Q + 3.461 * 22 + 13.390 * 3.5 \\ P = -58.7982 + 2.9367 * Q + 2.9585 * 23 \end{cases}$$

$$\begin{cases} P = 111.579 - 2.671 * Q \\ P = 9.2473 + 2.9367 * Q \end{cases}$$

	Q	P
structural	18.24843	62.83745
reduced-form	18.26010	62.81425

The equilibrium values obtained from the structural system are $Q^* = 18.248$ and $P^* = 62.837$. Using the reduced-form equations with $PS = 22$, $DI = 3.5$, and $PF = 23$, we obtain $\hat{Q} = 18.260$ and $\hat{P} = 62.814$.

The results from both approaches are extremely close, with only minor differences due to rounding and numerical approximation. This confirms that the reduced-form equations correctly reproduce the equilibrium of the structural model.

f. Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

- Demand

	OLS coef	p-value	IV coef	p-value
Q	0.1512	0.7642	-2.671	0.03154*
PS	1.3607	0.0303*	3.461	0.00458**
DI	12.3582	$3.48 \times 10^{-7}***$	13.390	$4.68 \times 10^{-5}***$

The coefficient of Q has the wrong (positive) sign and is statistically insignificant, indicating simultaneity bias due to endogenous price. IV corrects simultaneity bias and restores the law of demand.

- Supply

	OLS coef	p-value	IV coef	p-value
Q	2.6613	$5.42 \times 10^{-15}***$	2.9367	$1.32 \times 10^{-13}***$
PF	2.9217	$< 2 \times 10^{-16}***$	2.9585	$< 2 \times 10^{-16}***$

All coefficients have correct signs and are highly statistically significant. Supply equation is relatively stable across methods.

The OLS estimates of the demand equation are biased, as shown by the incorrect positive sign on quantity and its insignificance. This reflects simultaneity bias from endogenous price. In contrast, the supply equation shows correct signs and strong significance under both OLS and IV.

Compared with part (b), IV corrects the bias in demand and is consistent with economic theory, while supply estimates are relatively stable across methods.